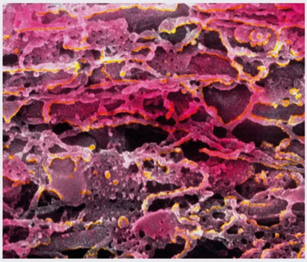
MEMBRANES OF ORGANELLES

Membranes divide the cytoplasm into sections and control the passage of materials between these regions, act as attachment points and storage areas, and shape channels along which substances move.

GOLGI COMPLEX
Within the membranous sacs of the Golgi complex, protein from the endoplasmic reticulum is processed.

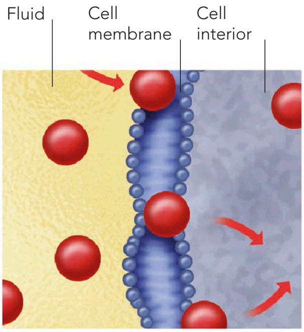


ENDOPLASMIC RETICULUM (ER)
A series of highly folded and curved ER membranes usually encloses one continuous labyrinthine space.

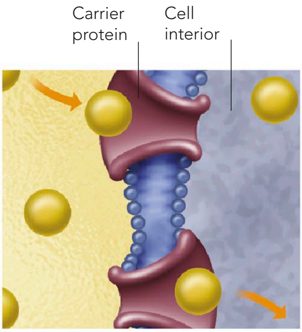
MITOCHONDRION
The inner membrane is folded to increase the area for releasing energy.

TRANSPORT

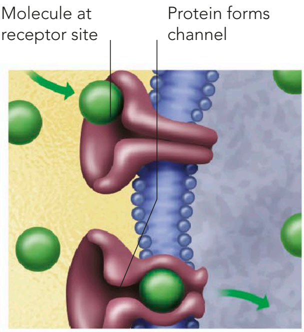
The transfer of materials through the cell membrane occurs by one of three processes. Small molecules, such as water, oxygen, and carbon dioxide, cross the membrane by diffusion. Molecules that cannot cross the phospholipid layer must cross by facilitated diffusion. When substances (such as minerals and nutrients) are lower in concentration on the outside of the cell than on the inside, they can only be conveyed into the cell by active transport, which requires energy.



DIFFUSION
Many molecules naturally move from an area where they are at high concentration to one in which their concentration is lower.



FACILITATED DIFFUSION
A carrier protein binds with a specific molecule outside the cell, then changes shape and ejects the molecule into the cell.



ACTIVE TRANSPORT
Molecules bind to a receptor site on the cell membrane, triggering a protein to change into a channel through which molecules travel.